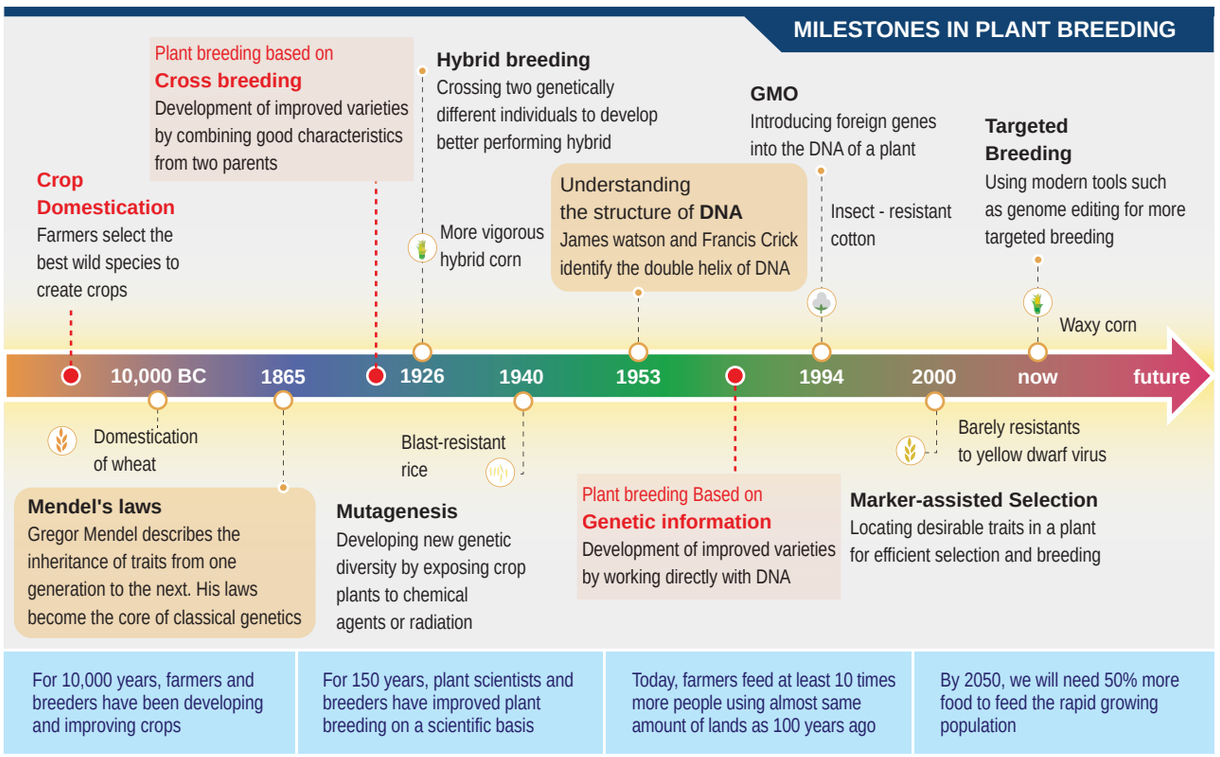


Figure 9.10 : Milestones in Plant Breeding

Biofortification – breeding crops with higher levels of vitamins and minerals or higher protein and healthier fats – is the most practical means to improve public health.

Breeding for improved nutritional quality is undertaken with the objectives of improving

- Protein content and quality
- Oil content and quality
- Vitamin content and
- Micronutrient and mineral content

In 2000, maize hybrids that had twice the amount of amino acids, lysine and tryptophan, compared to existing maize hybrids were developed. Wheat variety, Atlas 66 having a high protein content, has been used a donor for improving cultivated wheat. It has been possible to develop an iron fortified rice variety containing over five times as much iron as in commonly consumed varieties.

The Indian Agricultural Research Institute, New Delhi has also released several vegetable crops that are rich in vitamins and minerals, example: vitamin A enriched carrots, spinach, pumpkin; vitamin C enriched bitter gourd, bathua, mustard, tomato; iron and calcium enriched spinach and protein enriched beans – broad, lablab, French garden peas.

Sugar cane: *Saccharum bareri* was originally grown in North India, but had poor sugar content and yield. Tropical canes grown in South India *Saccharum officinarum* had thicker stems and higher sugar content but did not grow well in North India. These two species were successfully crossed to get sugar cane varieties combining the desirable qualities of high yield, thick stems, high sugar and ability to grow in the sugarcane areas of North India.

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